

PRIYANSH TOLAMBIYA

Bhilwara, Rajasthan

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Education

National Institute of Technology, Surat

B.Tech in Artificial Intelligence (3rd Year)

Gujarat

2024 – Present

Relevant Coursework: Machine Learning, Artificial Intelligence, Development and Operations, Data Structures and Algorithms, Operating Systems, Database Management Systems

Shree Mahesh Public School

Class 12 (CBSE)

Bhilwara, Rajasthan

2024

JEE Mains 2024: 98.99%ile — 88.2% in 10+2

Technical Skills

- **Programming Languages:** C, C++, Python, Java
- **Web Technologies:** HTML5, CSS3, Tailwind CSS, Bootstrap, JavaScript, TypeScript, React.js, Vue.js, Next.js, Node.js, Express.js, Nest.js, CI/CD
- **Machine Learning:** PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, NumPy, Pandas, LightGBM, XGBoost, CatBoost, LangChain
- **Databases:** MongoDB, MySQL, PostgreSQL
- **Developer Tools:** Git, GitHub, Docker, Kubernetes, AWS (EC2), FastAPI, REST APIs, PostgreSQL (pgvector), Matplotlib
- **Coding Profiles:** Solved 500+ DSA problems on LeetCode; achieved max rating of 1349 on Codeforces.

Projects

- **Intelligent Descriptive Search System for CCTV** GitHub
Python, PyTorch, YOLO, BoT-SORT, SigLIP, ReID, PostgreSQL (pgvector), FastAPI, Next.js, Docker
Built an end-to-end AI video analytics system enabling natural-language search across multi-camera CCTV footage; developed a video ingestion pipeline with object detection, tracking, attribute extraction, and embedding generation; implemented semantic search using pgvector with metadata filtering; designed cross-camera object tracing using re-identification and trajectory matching with a FastAPI backend and Next.js interface.
- **EnergyCast – AI-powered Energy Forecasting Platform** GitHub
Python, FastAPI, Next.js, TypeScript, LightGBM, XGBoost, CatBoost, Scikit-learn
Developed an AI-powered energy forecasting platform combining time-series ML, online learning, and explainable AI; built a full-stack dashboard for real-time and historical energy visualization; improved forecasting using ensemble models with online residual correction; integrated Gemini-powered explainability for actionable operator insights.
- **MindSettler – Mental Health Platform** Live Demo
Next.js, React, PostgreSQL, Prisma, LangChain, Google Gemini, Razorpay, Cloudinary
Built a full-stack mental health platform with AI chatbot, counselor booking, mood tracking, and journaling; developed secure REST APIs using NextAuth.js and Prisma; integrated LangChain with Gemini, Razorpay payments, and Cloudinary media management.
- **Movie Recommendation System** GitHub
Python, Pandas, NumPy, Scikit-learn
Developed a content-based recommendation engine using cosine similarity; performed data preprocessing, feature engineering, and vectorization to generate personalized movie recommendations.
- **Face Mask Detection System** GitHub
Python, OpenCV, TensorFlow/Keras, NumPy
Built a real-time CNN-based face mask detection system using OpenCV and TensorFlow/Keras; implemented face detection, model training, and real-time inference for mask compliance monitoring.
- **Wine Quality Prediction System** GitHub
Python, Pandas, NumPy, Scikit-learn, Matplotlib
Developed supervised learning models for wine quality prediction; performed preprocessing, feature engineering, normalization, exploratory data analysis, and model evaluation using standard classification metrics.

Interests

- Artificial Intelligence, Machine Learning, Deep Learning, Development and Operations, Competitive Programming, Data Structures and Algorithms.